

# Archit Mukherjee

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## Education

### Bachelor of Technology, Vellore Institute of Technology, Chennai, India

Sep 2020 - May 2024

CGPA: 9.43/10

- Major: **Electronics and Communication**
- Key Coursework: Neural Networks and Fuzzy Control, Digital Image Processing, Probability Theory and Random Processes, Information Theory and Coding, Embedded System Design

### Higher Secondary Education, Council for Indian School Certificate Examinations

Year 2020

Grade: 93.75%

- Major: **Computer Science**

### Secondary Education, Council for Indian School Certificate Examinations

Year 2018

Grade: 92.2%

- Major: **Science**

## Research Experience

### Singapore Management University

Singapore

Research Intern - Advisor: **Dr Archan Misra**

Dec 2023 - Jun 2023

- Developed a multimodal AI model designed to teach Geometry to primary and secondary students, enhancing their learning experience through interactive and adaptive methods.
- Utilized the DAVIS 346 Neuromorphic Event Camera for precise motion classification, focusing on detecting subtle variations in movement patterns.

### Samsung R&D Institute India, Bangalore

Remote

Samsung PRISM Research Intern - Mentor: **Ramesh Kumar L**

Feb 2023 - Oct 2023

- Developed novel approach of using Generative Adversarial Networks to convert flash photographs into ambient photographs by removing flash artefacts.
- Completion certificate: <https://www.samsungprism.com/Profile/Document/323VIS01VITC31272.pdf>

## Publications

### Posters

[Poster] **Archit Mukherjee**, Dulanga Weerakoon, Alok Prakash, Thivya Kandappu, Archan Misra, "Exploring AI-enabled Interactive Tutor for Teaching Geometry", **MIT-Singapore AI Symposium**, 2024

July 2024

Poster Link: <https://drive.google.com/file/d/1NsoaDDZlypS0IB4VNqquRntQRYKMnEBW/view?usp=sharing>

### Patents

[Design Patent Granted] VIT Chennai, Dr Rohith G, **Archit Mukherjee**, Tashmoy Ghosh, Sriram Kannan, Sarath Prathap. "Autonomous Mobility Vehicle for Hospital Sterile Instrument Logistics and Waste Collection during Surgeries". IN Design No.: 404191-001

Jan 2024

## Projects

### Neuromorphic Knitted Wearables (Ongoing)

2024

- Worked on developing a tinySNN model for low-power anomaly detection in 1D spike domain signals generated by neuromorphic knitted wearables.
- Explored various types of encoding schemes like rate coding, temporal coding, delta modulation, and loss functions like rate loss, count loss to improve responsiveness and accuracy.

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Automated Guided Vehicle using Visual Slam</b>                                                                                                                                                                                                                                                                                                                                                                                                    | 2023 |
| <ul style="list-style-type: none"> <li>• Built an AGV that maps and mobilizes in the given environment with the aid of an overhead camera that substitutes the expensive LiDARs.</li> <li>• The map was created with YOLO Algorithm; the path was planned with A* Algorithm and commands were given through ROS as middleware.</li> <li>• The AGV model used a Raspberry Pi to run the ROS server and Arduino Mega to control the motors.</li> </ul> |      |
| <b>IoT-based Smart Home</b>                                                                                                                                                                                                                                                                                                                                                                                                                          | 2022 |
| <ul style="list-style-type: none"> <li>• A smart home with custom lighting that can be controlled through a web app, with individual profiles with personalized lighting.</li> <li>• Implemented frontend with HTML/CSS and backend with Django.</li> </ul>                                                                                                                                                                                          |      |
| <b>IoT-based Smart Gardening</b>                                                                                                                                                                                                                                                                                                                                                                                                                     | 2022 |
| <ul style="list-style-type: none"> <li>• An automated plant watering and monitoring system with real-time image capturing using ESP32 Camera and ESP8266</li> <li>• Made a frontend with HTML/CSS.</li> </ul>                                                                                                                                                                                                                                        |      |
| <b>Android Controlled RC Vehicle using 8051 Microcontroller</b>                                                                                                                                                                                                                                                                                                                                                                                      | 2022 |
| <ul style="list-style-type: none"> <li>• Built an RC car controlled from an android device using Bluetooth.</li> <li>• Coded the 8051 Microcontroller in ASM to serially receive instructions from the HC-05 Bluetooth module.</li> </ul>                                                                                                                                                                                                            |      |
| <b>Automatic Vacuum Cleaning Bot using Arduino UNO</b>                                                                                                                                                                                                                                                                                                                                                                                               | 2021 |
| <ul style="list-style-type: none"> <li>• Built automatic vacuum cleaning bot using Arduino UNO, HC-SR04, and L293D Motor Driver.</li> <li>• Used the ultrasonic and IR sensors to detect obstacles and navigate accordingly.</li> </ul>                                                                                                                                                                                                              |      |

## Awards

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### 6th rank out of 229 students in the BTech ECE class of 2024 at VIT Chennai

Certificate Link: [https://drive.google.com/file/d/1XpoW9KQWZI3j\\_CSmETf34C1USUu0uZNg/view?usp=sharing](https://drive.google.com/file/d/1XpoW9KQWZI3j_CSmETf34C1USUu0uZNg/view?usp=sharing)

### Certificate of Excellence in Samsung PRISM Project

Certificate Link: <https://www.samsungprism.com/Profile/Document/223VIS01VITC31272.pdf>

### 6th Rank out of 229 in B.Tech. ECE in the fourth year (2023-24) at VIT Chennai

Certificate Link: <https://drive.google.com/file/d/1DPUN5E5tqU46ytMz9WnKOLYHZyGOj1uv/view?usp=sharing>

### 2nd Rank out of 229 in B.Tech. ECE in the second year (2021-22) at VIT Chennai

Certificate Link: <https://drive.google.com/file/d/1MYBiQ5qKlAGveGhr-BmJautrWVBBJTE3/view?usp=sharing>

### 7th Rank out of 229 in B.Tech. ECE in the first year (2020-21) at VIT Chennai

Certificate Link: [https://drive.google.com/file/d/1MYgDhxQo9T\\_ULX1m-j7epRo9pQMxWfYi/view?usp=sharing](https://drive.google.com/file/d/1MYgDhxQo9T_ULX1m-j7epRo9pQMxWfYi/view?usp=sharing)

### Innovation Award in International Rover Design Challenge 2021

Designed the Communication System of the innovative Assistive Bot for Mars Exploration

Certificate Link: [https://drive.google.com/file/d/1pk\\_xbjxez9r7-fljoQcTCrbyHuKsAj5U/view?usp=sharing](https://drive.google.com/file/d/1pk_xbjxez9r7-fljoQcTCrbyHuKsAj5U/view?usp=sharing)

## Skills

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**Languages:** Python, MATLAB, C, C++, Embedded C, Assembly Language, Core Java, R

**Technologies:** PyTorch, SNN Torch, ROS, Cadence Virtuoso, Autodesk Eagle

## Position of Responsibility

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### Captain of Team Technocrats Robotics

Sep 2022 - Jun 2023

A robotics team of VIT Chennai University focused on competitions like Robocon, IRDC, IRC and URC.

- Worked on designing the communication system of rover in IRDC 2021
- Designed PCB for rover power distribution considering safety measures and back EMF from motors as the Electrical head.
- As the captain, revitalized the team after Covid-19 break to start working for competitions by: eliminating inactive members; recruiting members through social media marketing, desk marketing, workshops, showcasing older projects; selection through quiz and interview and new component procurement.

## Certifications

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### Deep Learning Specialization

July 2022

A series of courses on DL and NN taught by Andrew Ng of DeepLearning.AI

- Strategies for error reduction in ML systems, and apply end-to-end, transfer, and multi-task learning
- CNN for visual detection and recognition tasks, use neural style transfer to generate art, Semantic Segmentation
- RNN, GRUs and LSTMs, apply RNNs to character-level language modeling, NLP and Word Embeddings
- Certificate Link: <https://coursera.org/share/1f31cc470503c141c37dd4d6095af39d>

### Data Analytics

May 2022 - July  
2022

Externship program on Applied Data Science powered by IBM

- Built a Machine Learning model using Scikit-learn to predict possibility of flood based on the rainfall
- Designed a web application using HTML, CSS and Bootstrap and deployed it using IBM Cloud
- Certificate Link: <https://drive.google.com/file/d/1f5BEAyxM8JDxw3IK3D5aQp09it0eNxEZ/view?usp=sharing>